

Customer: Suzhou PTC Optical Instrument Co Ltd **Date:** 11-Apr-23 **Time:** 10:00

Participant From Maruwa	Participant From Organizer	Report To
1. Mr Siva 2. Mr Ronny 3. 4 5.	1. Mr Bo Hong 2. Mr Edwin 3. 4 5.	1. Datuk Anthony & Mr Sasaki 2. Mr Sinna & Mr Nazli 3. Mr Okui & Mr Kitauji 4 Mr Sathia & Ms Pang 5. Mr Mori & Mr Ogawa

Agenda

Meeting started with presenting the company profile (Automation company recommended by Ferrotec)
Brief detail about this company

1. Total workforce : 320paxs
2. Company initiated in 2007
3. Supporting Apple since Iphone 8 onwards (Touch screen inspection)
4. Working with Ferrotec on various AOI inspection especially at mid process of DBC, DPC & AMB
5. Next project - Ferrotec Malaysia plant setting up of AOI machines

Below is the summary of the supplier capability

AUTOMATION SUMMARY - SUPPLIER CAPABILITY & CAPACITY							
No	MACHINE TYPE	CUSTOMER	APPLICATION	CAPABILITY	CAPACITY	AREA OF SIMILARITY TO MARUWA	LEVEL OF ACCEPTANCE
1	AOI - AUTO VISUAL INSPECTION	FERROTEC	Inspection for a. DBC b. DPC c. AMB - Pithole Inspection	Process step a. Visual inspection - Top b. Thickness check - 5point (4 corners + Centre) c. Product Flip d. Bottom Visual Inspection => Visual defects i. Crack (White) ii. Stain iii. Chipping iv. Pithole(pinhole in the middle) e. Binning (OK + Defects)	1st Generation model : 600pcs/hr : 7200pcs/shift 2nd Generation model 1000pcs/hr : 12,000pcs/shift (Excluded Thickness check)	TPH Visual inspection	90%
2	AOI - Green Sheet inspection	Substrate supplier - Jiuhou	Inspection for green sheet surface	Process Steps a. Visual inspection - Top b. Thickness check - Scanning (Overall surface) c. Product Flip d. Bottom Visual Inspection => Visual defects i. Scratches ii. Boro iii. Dent iv. Boro (Cutting edge) e. Recycle	No data	TPH Press Greensheet inspection (online)	90%
3	AOI - Magnet bar inspection	Apple	Inspection Ferrite bar inspection	To check on the ferrite bar visual condition	No data	FM - BB type - Slicing products	50%
4	AOI - Printing screen inspection	Substrate users	Printing screen dimension & visual	To check on the printing pitch & dimension	-	Not related	-
5	-	-	AOI-Inspection	Shown the film mnspection defects a. Stain/ Ceramic piece b. Wave => 1st Review : capable to do identify the defects	-	Film Inspection	-

General Info

a. Machine fabrication leadtime : 60days

Meeting adjourned at 2:30pm

Conclusion & Opinion

As observe, the supplier is capable & experience in the AOI implementation. They also have machines which is in running condition with consistent of GR&R on the output results. This supplier have being engaged by high end customers such as Ferrotec, Apple & etc. The comparison & comments will be shared to Malaysia & Japan automation team for further review.



Customer Visit Report

MM-IN-RD-9100-P06-04
Rev: 1

Customer: KUNSHAN YIYUCHENG INTELLIGENT TECHNOLOGY		Date: 11-Apr-23	Time: 15:00
Participant From Maruwa		Participant From Organizer	Report To
1. Mr Siva	Maruwa Malaysa	1. Mr Eric	1. Datuk Anthony & Mr Sasaki
2. Mr Ronny	Maruwa Shanghai	2.	2. Mr Sinna & Mr Nazli
3.		3.	3. Mr Okui & Mr Kitauji
4		4	4 Mr Sathia & Ms Pang
5.		5.	5. Mr Mori & Mr Ogawa

Agenda

Meeting started with presenting on the machine capability

Brief detail about this company

- 1. Mainly working with Substrate makers in China
- 2. Main customers are : Jiuhou, Sankang, Uniroyal etc

Below is the summary of the supplier capability

AUTOMATION SUMMARY - SUPPLIER CAPABILITY & CAPACITY

No	MACHINE TYPE	CUSTOMER	APPLICATION	CAPABILITY	CAPACITY	AREA OF SIMILARITY TO MARUWA	LEVEL OF ACCEPTANCE
1	Auto Warp	Jiuhou	Warp inspection	Carry out warp inspection from 0.47 ~ 0.5 Spec : +0.10 Machine running condition	Total 3 Lane 1 min : 50pcs (3 lane) 36,000pcs/shift	TFC Gating	90%
2	Auto Taping	Jiuhou	Taping before 2nd Firing	To carry out tapping process before proceeding for 2nd firing => Bundle size : Preset (10pcs ~ 20pcs)	8 bundle/min (10pcs/bundle)	TFC Honing Taping TFC Warp Taping	90%

Meeting adjourned at 4:30pm

Conclusion & Opinion

This supplier is on evaluation for Jiuhou for auto visual on substrate and lack of capability to do the AOI for visual both with slit or blank substrate. Currently commisioning for Warp inspection & taping process for Jiuhou in Kunshan.

CHINA AUTOMATION – SUPPLIER PERFORMANCE / CAPABILITY

チャイナオートメーション-サプライヤーのパ
フォーマンス/機能

Date: 11th Apr 2023

CONFIDENTIAL

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